



GIRRAWEEEN HIGH SCHOOL

TRIAL EXAMINATION

2018

MATHEMATICS

*Time allowed - Three hours
(Plus 5 minutes reading time)*

General Instructions

- Reading time-5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In Questions 11-15, show relevant mathematical reasoning and/or calculations.

Total marks : **Part A- 10 marks**
100

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Part B - 90 marks

- Attempt Questions 11-15
- Allow about 2 hours and 45 minutes for this section

PART A: Questions 1 – 10 (Multiple Choice) *Write the letter corresponding to the correct answer on your working pages.*

Question 1

The condition for the quadratic equation $3x^2 - 12x + k = 0$ to have real roots is

- A) $k \leq 36$ B) $k \geq 36$ C) $k \leq 12$ D) $k \geq 12$

Question 2

A yacht sailed directly from A to B on a bearing of $196^\circ T$. To sail from B back to A , the bearing should be:

- A) $016^\circ T$ B) $074^\circ T$ C) $164^\circ T$ D) $196^\circ T$

Question 3

$\log_3 15 + \log_3 18 - \log_3 10$ evaluates to :

- A) 1 B) 2 C) 3 D) 0

Question 4

What is the size of the angle subtended at the centre of a circle with diameter 7cm by an arc 3cm long?

- A) 25° B) 49° C) 67° D) 133°

Question 5

What is the limiting sum of the series $45 - 15 + 5 - \frac{5}{3} + \dots\dots\dots$?

- A) $\frac{135}{2}$ B) $\frac{45}{4}$ C) $\frac{135}{4}$ D) $\frac{-45}{2}$

Question 6

The area between the curve $y = \frac{1}{x}$, the x axis and the lines $x = 1$ and $x = b$ is equal to 2 square units. The value of b is

- A) e B) e^2 C) $2e$ D) 3

Question 7

What is the period of the function $y = 4\sin\left(\frac{x}{3}\right)$?

- A) 6π B) 4 c) $\frac{2\pi}{3}$ D) $\frac{1}{4}$

Question 8.

The first term of an arithmetic progression is 3, and eleventh term is 23. The n^{th} term is

- A) $T_n = 3 + 23(n-1)$ B). $T_n = 23 + 10(n-1)$ C). $T_n = 3 + 2(n-1)$ D) $T_n = 3 + 10(n-1)$

Question 9.

If $\log_{10} 7 = a$ then $\log_{10}\left(\frac{1}{70}\right)$ is equal to

- A) $-(1+a)$ B) $(1+a)^{-1}$ C) $\frac{a}{10}$ D) $\frac{1}{10a}$

Question 10.

The roots of the quadratic equation $gx^2 - x + h = 0$ are -1 and 3. The value of h is

- A) -6 B) -3 C) $\frac{-3}{2}$ D) 2

PART B: Show all necessary working on your answer pages.

Question 11(18 Marks).

Marks

a) Find the value of $e^{2\pi}$ correct to 2 significant figures.

2

b) Factorise $3x^2 + x - 2$.

2

c) If $\frac{20\sqrt{18}}{16\sqrt{24}} = a\sqrt{3}$ find the value of a .

2

d) Solve $2\log_3 5 - \log_3 x = 2$

2

e) Given that $\tan \theta = \frac{7}{8}$ and $\cos \theta < 0$, find the exact value of $\operatorname{cosec} \theta$.

2

f) Differentiate:

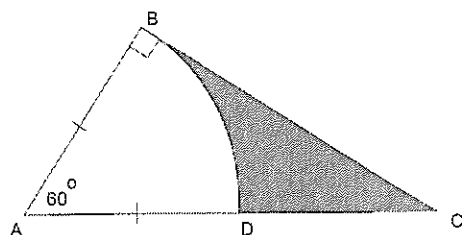
i) $y = (e^x + 2)^8$

2

ii) $y = \frac{x-1}{\cos x}$

3

g) In the diagram $\angle B = 90^\circ$, $\angle A = 60^\circ$ and $AB = AD = 10m$. BD is an arc of the circle with centre A .



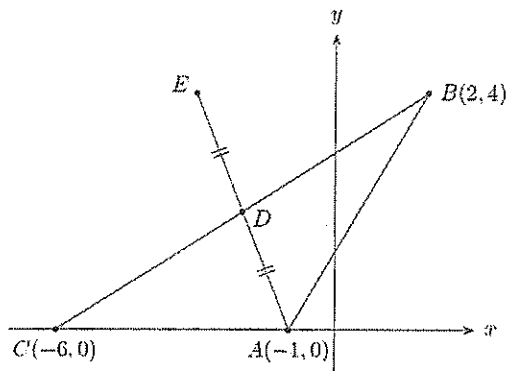
3

Calculate the shaded area in exact form.

Question 12 (18 Marks).

a) In the diagram below, A, B, C and D are the points $(-1,0), (2,4), (-6,0), (-2,2)$ respectively.

D is also the midpoint of AE .



In the diagram above,

- i) Find the length of the interval AB . 1
- ii) Find the equation of the circle with centre at B which passes through the point A . 1
- iii) Find the size of $\angle CAB$ to the nearest degree. 2
- iv) Find the midpoint of BC . 1
- v) Show that the equation of the line BC is $x - 2y + 6 = 0$ 1
- vi) Find the perpendicular distance of A from the line BC in simplest exact form. 2
- vii) What type of quadrilateral is $ABEC$? Give reasons for your answer. 2

b) The roots of the equation $x^2 - 8x + 5 = 0$ are α and β .

Find the value of $(\alpha - \beta)^2$. 2

c) The rate at which water flows into a tank is given by

$$\frac{dV}{dt} = \frac{2t}{1+t^2},$$

Where V is the amount of water in the tank in litres and t is the time in seconds.

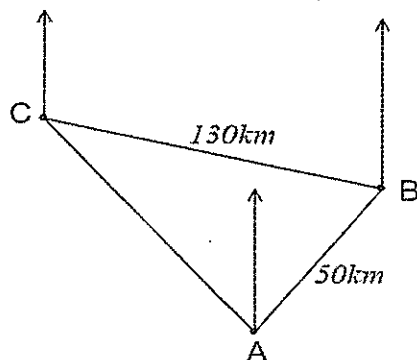
Initially the tank is empty. Find the exact amount of water in the tank after

10 seconds. 3

d) A geometric series has first term a and limiting sum 2. Find all possible values for a . 3

Question 13 (18 Marks).

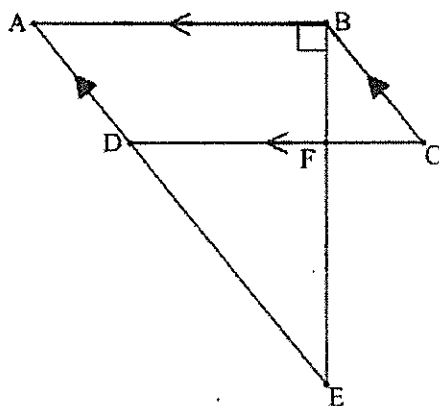
- a) A ship sails 50km from port A to port B on a bearing of 063° . It then sails 130km from port B to port C on a bearing of 296° .



- i) Show (with working) $\angle ABC = 53^\circ$. 2
 ii) Find AC to the nearest kilometre. 3

- b) In the following diagram, $ABCD$ is a parallelogram where $FB \perp AB$. 3

- i) Prove that $\triangle CBF \parallel \triangle AEB$
 ii) If $CF = 3\text{cm}$, $BC = 7\text{cm}$ and $AE = 15\text{cm}$, find the length of AB . 2



NOT TO SCALE

- c) Find the equation of the normal to the curve $y = 1 + \ln 2x$ at the point $\left(\frac{e}{2}, 2\right)$. 3

- d) Solve $3\sin^2 \theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$ (Give your answer to 2 decimal places in radians) 2

- e) By letting $m = t^{\frac{1}{3}}$, or otherwise, solve $t^{\frac{2}{3}} + t^{\frac{1}{3}} - 6 = 0$ 3

Question 14 (18 marks).

a) Consider the function $f(x) = x^3 - x^2 - 5x + 1$.

i) Find the coordinates of the stationary points of the curve $y = f(x)$ and

determine their nature.

2

ii) Find any points of inflexion.

1

iii) Sketch the curve $y = f(x)$ for $-2 \leq x \leq 2$ clearly indicating the end points.

(You need not find the x -intercepts.

2

iv) For what values of x is the curve $y = f(x)$ decreasing but concave up?

1

b) i) If $y = \sqrt{\sin x}$, complete the table below.

x	0	0.5	1	1.5	2
y					0.95

1

ii) Hence evaluate $\int_0^2 \sqrt{\sin x} \, dx$ using Simpson's rule with 5 function values.

2

c) Find

$\alpha) \int 6e^{\frac{x}{2}} dx$

1

$\beta) \int \frac{x}{1-x^2} dx$

1

$\gamma) \int_0^{\frac{\pi}{6}} (1 - \sec^2 2x) dx$

2

d) R is the region bounded by the y -axis, the x -axis, the line $x = \frac{\pi}{2}$ and the curve

$y = \sqrt{1 + \sin x}$. Find the volume of the solid formed when the region R is rotated about the x -axis.

3

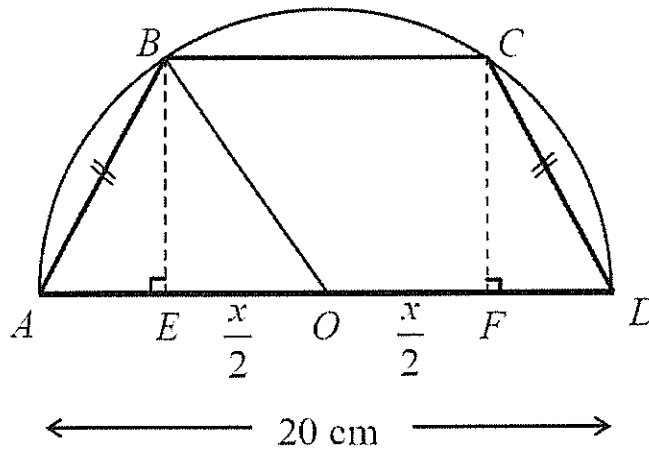
e) Consider the function $f(x) = kx^2 - (3k-4)x + k$.

Show that $f(x)$ is positive definite for $\frac{4}{5} < k < 4$.

2

Question 15 (18 marks).

a) An isosceles trapezium $ABCD$ is drawn with its vertices on a semicircle centre O and diameter 20 cm. Perpendiculars BE and CF are drawn to meet the diameter AD as illustrated in the diagram below.



i) If $EO = OF = \frac{x}{2}$, show that $BE = \frac{1}{2}\sqrt{400 - x^2}$ 2

ii) Show that the area of the trapezium $ABCD$ is given by $\frac{1}{4}(x + 20)\sqrt{400 - x^2}$. 2

iii) Hence find the length of BC so that the area of the trapezium $ABCD$ is a maximum. 3

b) A particle P is at the origin at time $t = 0$ and moves so that its velocity for $t \geq 0$ is given by $v = \frac{1}{t+1}$.

i) What is the acceleration of P when $t = 1$? 2

ii) What is the displacement of x of P from the origin when $t = 1$? 2

Question 15 continues on the next page

- c) The population P of a town is growing at a rate proportional to its size at any time, so that $\frac{dP}{dt} = kP$, for some constant k . At the beginning of 2010 (where $t=0$ and time is in years) the town's population was 23000 and at the beginning of 2016 its population had grown to 28000.
- i) Show that $P = Ae^{kt}$ satisfies the equation $\frac{dP}{dt} = kP$. 1
- ii) Find the value of A . 1
- iii) Find the value of k . 2
- iv) Estimate, to the nearest hundred, what the population will be at the beginning of 2025. 1
- v) During which year will the population be double the size it was at the beginning of 2010. 2

End of examination!!!

1

Part A

1) C

2) A

3) C

4) B

5) C

6) B

7) A

8) C

9) A

10) C

Question 11

a) $e^{2\pi} = 535.49 = 540$ to 2 sig. figures

b) $3x^2 + x - 2 = (3x - 2)(x + 1)$

c) $\frac{20\sqrt{18}}{16\sqrt{24}} = \frac{5\sqrt{3}}{4\sqrt{4}} = \frac{5\sqrt{3}}{8} \therefore a = \frac{5}{8}$

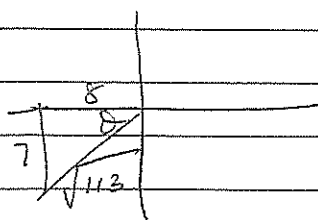
d) $2\log_3 5 - \log_3 x = 2$

$\log_3 5^2 - \log_3 x = 2$

$\log_3 \frac{25}{x} = 2 \therefore \frac{25}{x} = 3^2 = 9$

$\therefore x = \frac{25}{9}$

e)



$\tan \theta = +ve, \cos \theta < 0$

$\therefore$ 3rd quadrant

$\therefore \sin \theta = \frac{-7}{\sqrt{113}}$

$\therefore \operatorname{cosec} \theta = -\frac{\sqrt{113}}{7}$

f) (i) $y = (e^x + 2)^8$

$\frac{dy}{dx} = 8(e^x + 2)^7 \times e^x = 8e^x(e^x + 2)^7$

$$(f)(ii) \quad y = \frac{x-1}{\cos x}$$

$$u = x-1 \quad \left| \quad v = \cos x \right.$$

$$u' = 1 \quad \left| \quad v' = -\sin x \right.$$

$$\frac{dy}{dx} = \frac{v u' - u v'}{v^2} = \frac{\cos x + (x-1)\sin x}{\cos^2 x}$$

$$= \frac{\cos x + x \sin x - \sin x}{\cos^2 x}$$

9) In ΔABC , $\tan 60^\circ = \frac{BC}{10} = \sqrt{3} \quad \therefore BC = 10\sqrt{3}$

$$\text{Area of } \Delta ABC = \frac{1}{2} \times 10 \times 10\sqrt{3} = 50\sqrt{3} \text{ m}^2$$

$$\text{Area of sector ABD} = \frac{1}{2} \times 10 \times 10 \times \frac{\pi}{3} = \frac{50\pi}{3} \text{ m}^2$$

$$\therefore \text{shaded area} = 50\sqrt{3} - \frac{50\pi}{3}$$

$$= \frac{50}{3} [3\sqrt{3} - \pi] \text{ m}^2$$

Question 12

a) (i) $AB = \sqrt{(2+1)^2 + (4-0)^2} = 5$

(ii) $(x-2)^2 + (y-4)^2 = 25$

(iii) $m_{AB} = \tan \theta = \frac{4}{3} \quad \therefore \theta = \tan^{-1} \frac{4}{3} = 53.13^\circ$

$$\therefore \angle CAB = 180^\circ - 53.13^\circ = 126.87^\circ$$

(iv) $m = \left(\frac{2-6}{2}, \frac{4-0}{2} \right) = (-2, 2)$

(v) $m_{BC} = \frac{4-0}{2+6} = \frac{1}{2}$

~~B~~ $(-6, 0)$ $\therefore$ equation of BC $\Rightarrow$

$$y - 0 = \frac{1}{2}(x + 6) \Rightarrow x - 2y + 6 = 0$$

(vi) $(-1, 0)$ $x - 2y + 6 = 0$

$$\therefore \text{perpendicular distance} = \frac{|-1 - 2(0) + 6|}{\sqrt{1+4}}$$

$$= \frac{5}{\sqrt{5}} = \sqrt{5}$$

(vii) As D is the mid point of AE as well,
 ABEC is a parallelogram.
 (diagonals bisect each other)

But as $AC = 5$ as well as $AB = 5$,
 ABEC is a rhombus.
 (adjacent sides are equal and has properties of a parallelogram)

b) $x^2 - 8x + 5 = 0$

$\alpha + \beta = 8$ and $\alpha\beta = 5$

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$= 64 - 20 = 44$$

c) $\frac{dv}{dt} = \frac{2t}{1+t^2}$ when $t=0$, $v=0$

$$\therefore v = \int \frac{2t}{1+t^2} dt = \ln[1+t^2] + C$$

when $t=0$, $v=0 \Rightarrow 0 = \ln 1 + C \Rightarrow C = 0$

$$\therefore v = \ln(1+100) = \ln(101)$$

d) $S_{\infty} = \frac{a}{1-r} = 2$

$$\frac{a}{2} = 1-r \Rightarrow r = 1 - \frac{a}{2}$$

For S_{∞} , $|r| < 1$ $\left|1 - \frac{a}{2}\right| < 1$

$$-1 < 1 - \frac{a}{2} < 1$$

$$-2 < 2 - a < 2$$

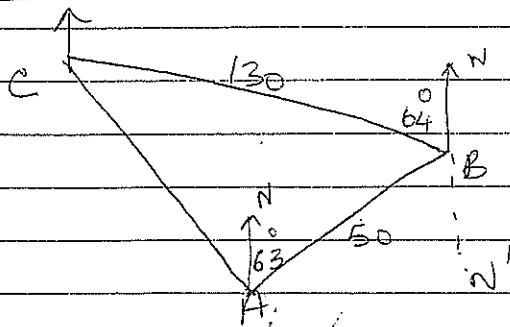
$$2 > a - 2 > -2$$

$$\Rightarrow 4 > a > 0$$

$$\therefore 0 < a < 4$$

Question 13

a) (i)



$$\angle ABN = 63^\circ \text{ (alternate } \angle \text{ s on } \parallel \text{ lines)}$$

$$\angle ABC = 180^\circ - (63 + 64) \quad \begin{matrix} \text{(} \angle \text{ s on a line} \\ \text{= } 180^\circ \text{)} \\ \text{(} \angle \text{ s on a straight line)} \end{matrix}$$

$$= 53^\circ$$

(ii)

$$AC^2 = AB^2 + BC^2 - 2AB \cdot BC \cos 53^\circ$$

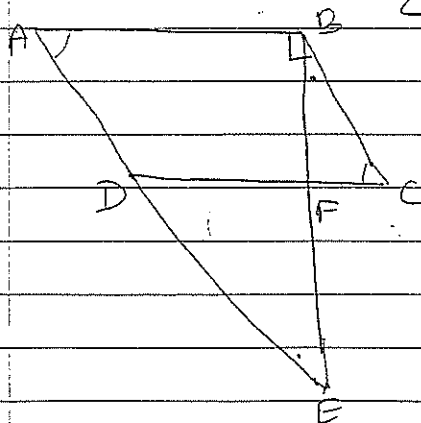
$$= 50^2 + 130^2 - 2 \times 50 \times 130 \cos 53^\circ$$

$$= 11576$$

$$AC = \sqrt{11576} = 108 \text{ km (nearest km)}$$

(b) (i)

In Δ s CBF, AEB



$\angle CBF = \angle BEA$ (alternate angles on $\parallel$ lines, $AE \parallel BC$)

$\angle BCF = \angle EAB$ (opposite angles of a parallelogram =)

$\therefore \Delta CBF \parallel \Delta AEB$ (equiangular)

(ii)

$$\frac{AB}{BF} = \frac{AE}{EC} \quad \therefore AB = \frac{AE \times CF}{BC}$$

(Ratio of matching sides) $= \frac{15 \times 3}{7} = 6\frac{3}{7} \text{ m}$

c)

$$y = 1 + \ln 2x$$

$$\text{at } \left(\frac{e}{2}, 2\right)$$

$$y' = \frac{1}{2x} \cdot 2 = \frac{1}{x}$$

$$\text{at } \left(\frac{e}{2}, 2\right), y' = \frac{2}{e}$$

$$\therefore y'_{\text{normal}} = -\frac{e}{2}$$

$\therefore$ equation of the normal:

$$y - 2 = -\frac{e}{2}\left(x - \frac{e}{2}\right)$$

$$2y - 4 = -ex + \frac{e^2}{2} \Rightarrow 4y - 8 = -2x + e^2$$

$$\therefore 2x + 4y - 8 - e^2 = 0$$

d)

$$3\sin^2 \theta = \sin \theta \quad \text{for } 0 \leq \theta \leq 2\pi$$

$$3\sin^2 \theta - \sin \theta = 0$$

$$\sin \theta (3\sin \theta - 1) = 0$$

$$\therefore \sin \theta = 0 \quad \text{or} \quad 3\sin \theta = 1$$

$$\sin \theta = \frac{1}{3}$$

$$\therefore \theta = 0, \pi, 2\pi \quad \text{or} \quad \theta = 0.34, 2.8$$

$$\therefore \theta = 0, 0.34, 2.8, \pi, 2\pi$$

e)

$$m = t^{1/3} \quad \therefore m^2 = \left(t^{1/3}\right)^2 = t^{2/3}$$

$$t^{2/3} + t^{1/3} - 6 = 0$$

$$\therefore m^2 + m - 6 = 0$$

$$(m+3)(m-2) = 0$$

$$\therefore m = -3 \quad \text{or} \quad m = 2$$

$$t^{1/3} = -3 \quad \text{or} \quad t^{1/3} = 2$$

$$\therefore t = -27$$

$$\therefore t = 8$$

Question 14

a) $f(x) = x^3 - x^2 - 5x + 1$

$$f'(x) = 3x^2 - 2x - 5$$

$$f''(x) = 6x - 2$$

$f'(x) = 0$ for stationary points.

$$\text{i.e. } 3x^2 - 2x - 5 = 0$$

$$(3x - 5)(x + 1) = 0$$

$$\therefore x = -1 \text{ or } \frac{5}{3}$$

If $x = -1$, $f''(-1) = 6(-1) - 2 < 0 \therefore$ maximum

$$\begin{aligned} f(-1) &= (-1)^3 - (-1)^2 - 5(-1) + 1 \\ &= -1 - 1 + 5 + 1 = 4 \end{aligned}$$

$\therefore$ a maximum at $(-1, 4)$.

If $x = \frac{5}{3}$, $f''(\frac{5}{3}) = 6(\frac{5}{3}) - 2 > 0 \therefore$ minimum

$$\begin{aligned} f(\frac{5}{3}) &= (\frac{5}{3})^3 - (\frac{5}{3})^2 - 5(\frac{5}{3}) + 1 \\ &= -5\frac{13}{27} \end{aligned}$$

$\therefore$ a minimum at $(\frac{5}{3}, -5\frac{13}{27})$

(ii) $\therefore$ for points of inflexion when $f''(x) = 0$
i.e. $6x - 2 = 0 \quad \therefore x = \frac{1}{3}$

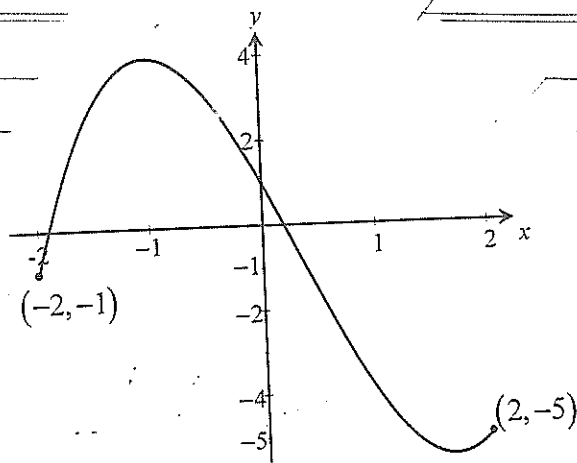
x	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$
$f''(x)$	-0.5	0	1

$\therefore$ concavity changes

$$\begin{aligned} f(\frac{1}{3}) &= (\frac{1}{3})^3 - (\frac{1}{3})^2 - 5(\frac{1}{3}) + 1 \\ &= -\frac{20}{27} \end{aligned}$$

$\therefore$ an inflexion at $(\frac{1}{3}, -\frac{20}{27})$

(iii)



(iv)

$y = f(x)$ is decreasing, but concave up when

$$\frac{1}{3} < x < \frac{5}{3}$$

b) (i)

$$y = \sqrt{\sin x}$$

x	0	0.5	1	1.5	2
y	0	0.69	0.92	1.0	0.95

$$(ii) \quad A = \frac{0.5}{3} [0 + 0.95 + 4(0.69 + 1) + 2(0.92)]$$

$$= 1.5916 \dots = 1.6$$

(iii)

$$(a) \quad \int 6e^{\frac{x}{2}} dx = 6 \cdot \frac{e^{\frac{x}{2}}}{\frac{1}{2}} = 12e^{\frac{x}{2}} + C$$

$$(b) \quad \int \frac{x}{1-x^2} dx = -\frac{1}{2} \int \frac{-2x}{1-x^2} dx = -\frac{1}{2} \ln|1-x^2| + C$$

$$\begin{aligned}
 (c) \quad \int_0^{\frac{\pi}{6}} (1 - \sec^2 2x) dx &= \left[x - \frac{\tan 2x}{2} \right]_0^{\frac{\pi}{6}} \\
 &= \left(\frac{\pi}{6} - \frac{1}{2} \tan \frac{\pi}{3} \right) - (0 - 0) \\
 &= \frac{\pi}{6} - \frac{\sqrt{3}}{2}
 \end{aligned}$$

Question 14 Continued

d)
$$V = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} (1 + \sin x) dx$$

$$= \pi \left[x - \cos x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left[\frac{\pi}{2} - \cos \frac{\pi}{2} - 0 + 1 \right]$$

$$= \pi \left[\frac{\pi}{2} + 1 \right] = \frac{\pi}{2} (\pi + 2) u^3$$

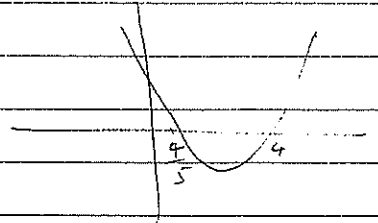
e)
$$f(x) = kx^2 - (3k-4)x + k$$

positive definite $\Rightarrow \Delta < 0$ and $a > 0$.

$$\Delta = (3k-4)^2 - 4k^2 = 9k^2 - 24k + 16 - 4k^2 \\ = 5k^2 - 24k + 16 < 0$$

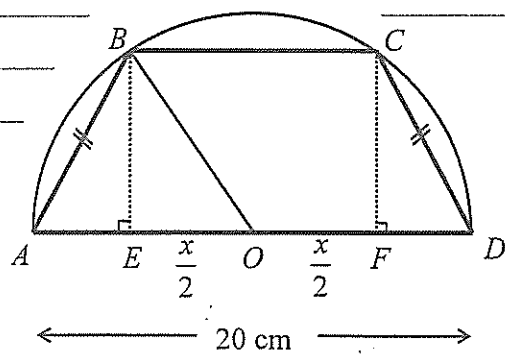
$$\Rightarrow (5k-4)(k-4) < 0$$

$$\therefore \frac{4}{5} < k < 4$$



Question 15

g)



diameter = 20 cm

∴ radius = 10 cm

(i) In $\triangle OBE$, $OB^2 = BE^2 + OE^2$ (Pythagoras' theorem)

$$\therefore 10^2 = BE^2 + \frac{x^2}{4} \Rightarrow BE^2 = 100 - \frac{x^2}{4} = \frac{1}{4}(400 - x^2)$$

$$\therefore BE = \frac{1}{2}\sqrt{400 - x^2} \quad (\text{length} \geq 0)$$

(ii)

$$A_{\text{Trap.}} = \frac{1}{2}h(a+b)$$

$$= \frac{1}{2} \cdot BE (BC + AD)$$

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot \sqrt{400 - x^2} \cdot (x + 20)$$

$$= \frac{1}{4}(x+20)\sqrt{400-x^2}$$

$$(iii) A = \frac{1}{4}(x+20)\sqrt{400-x^2} = \frac{1}{4}(x+20)(400-x^2)^{\frac{1}{2}}$$

$$\therefore \frac{dA}{dx} = \frac{1}{4} \cdot (x+20) \cdot \frac{1}{2}(400-x^2)^{-\frac{1}{2}}(-2x) + (400-x^2)^{\frac{1}{2}} \cdot \frac{1}{4}$$

$$= -\frac{x}{4}(x+20)(400-x^2)^{-\frac{1}{2}} + \frac{1}{4}(400-x^2)^{\frac{1}{2}}$$

$$= \frac{1}{4} \left[\frac{\sqrt{400-x^2} - \frac{x(x+20)}{\sqrt{400-x^2}}}{\sqrt{400-x^2}} \right]$$

$$\text{Maximum occurs} \Rightarrow \frac{1}{4} \left[\frac{400-x^2 - x^2 - 20x}{\sqrt{400-x^2}} \right] = 0$$

$$\text{when } \frac{dA}{dx} = 0$$

$$= \frac{1}{4} \frac{(2x^2 + 20x - 400)}{\sqrt{400-x^2}} = 0$$

$$2x^2 + 20x - 400 = 0$$

$$x^2 + 10x - 200 = 0$$

$$(x + 20)(x - 10) = 0$$

$$\therefore x = 10 \quad (x \geq 0)$$

x	9	10	11
$\frac{dA}{dx}$	0.8	0	-0.796

$\therefore$ The maximum occurs at when $x = 10$

$\therefore BC = 10 \text{ cm.}$

Question 15

b) (i) $V = \frac{1}{t+1} = \frac{dx}{dt}$

$$\therefore \ddot{x} = \frac{dv}{dt} = -(1+t)^{-2} = \frac{-1}{(1+t)^2} = -\frac{1}{4}$$

when $t = 1$

(ii) $\int dx = \int \frac{dt}{1+t} = \ln(t+1) + C$

$$\therefore x = \ln(t+1) + C$$

$$t=0, x=0 \Rightarrow 0 = \ln 1 + C \Rightarrow C=0$$

$$x = \ln(1+t) = \ln 2$$

when $t=$

Question 15 continued

(i) $P = Ae^{kt}$

$$\frac{dP}{dt} = k \cdot Ae^{kt} = kP$$

(ii) $P = Ae^{kt}$

In 2010, $t = 0$, $P = 23000$

$$23000 = Ae^0 = A$$

$$\therefore A = 23000$$

(iii)

In 2016, $t = 6$, $P = 28000$

$$28000 = 23000e^{6k}$$

$$\therefore k = \frac{1}{6} \ln\left(\frac{28}{23}\right)$$

$$\approx 0.032785 \dots$$

(iv)

$t = 15$, $P = ?$

$$P = 23000e^{15k}$$

$$\approx 37600 \text{ (to nearest hundred)}$$

(v)

$P = 46000$, $t = ?$

$$46000 = 23000e^{kt}$$

$$\Rightarrow 2 = e^{kt}$$

$$\therefore kt = \ln 2$$

$$\therefore t = \frac{1}{k} \ln 2$$

$$\approx 21.14$$

$\therefore$ doubles during the 22nd year.

i.e. 2031